Benchmarking Robustness of Vision Foundation Models for Prostate Cancer Relapse Prediction

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Introduction. Vision Foundation Models (VFMs) have shown promising results in field of histopathology, e.g. in cancer subtyping [2]. These models are supposed to be more robust against domain shifts than task-specific models, mainly achieved by pre-training on huge datasets from multiple organs. While many studies evaluate the VFMs on datasets from different clinics with unknown imaging protocols and patient cohorts [5], we analyse the robustness of different VFMs on our high-variance imaging dataset where domain shifts are intentionally introduced in the imaging protocol on the same patient cohort. This analysis shows that even large VFMs still show some problems in terms of robustness.

Materials and Methods. The benchmarking is based on 10,503 H&E-stained images of Tissue Microarray (TMA) spots and respective five-year follow-up on relapse-related events provided by the Institute of Pathology of the University Medical Center Hamburg-Eppendorf (UKE). The images come from 7,528 patients that underwent Radical Prostatectomy (RP). In addition to TMAs that were collected using the standard imaging protocol (*uke.first*) and were used for training, five subdatasets that differ in the tissue region (*uke.spot*: second TMA spot of a patient), the scanner (*uke.scanner*), the slicing (*uke.thin*: thinner, *uke.thick*: thicker) and the staining time (*uke.long*: longer) were collected.

To evaluate the robustness of VFMs, we decided to compare the following VFMs as they represent a diverse selection with respect to Vision Transformer (ViT) architecture and size of pre-training dataset: Kaiko-S, Kaiko-B, Kaiko-L [4], Virchow2 [6], UNI2-h [3] and H-optimus-1 [1]. They are evaluated as frozen encoders and extended with an attention-based Multiple Instance Learning (MIL) layer and a binary classification head to predict relapse.

Results and Discussion. Evaluating the VFMs on the subdataset *uke.first*, which was used for training, shows performance differences between the models with more parameters (Fig. 1, dashed line, second row) being superior to ones with less (Fig. 1, dashed line, first row). The best performance of 0.697 is achieved with the largest model H-optimus-1 (1.1B parameters).

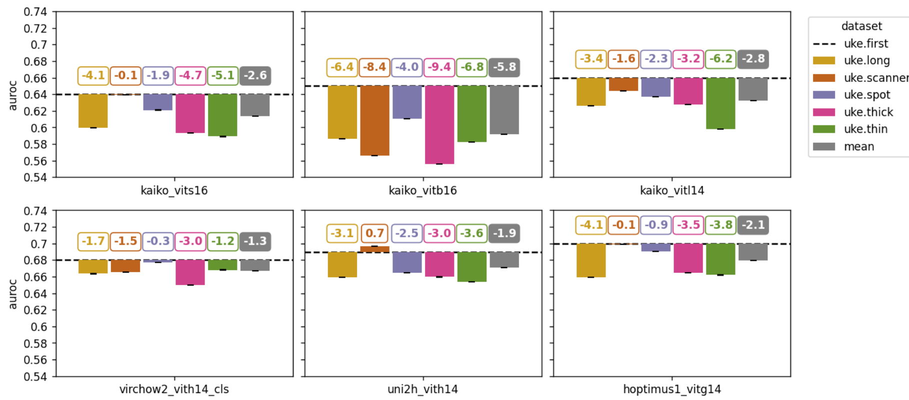

Fig. 1. Results of the robustness evaluation of vision foundation models.

The performance on the other subdataset drops on average by 1.3 - 5.8 percentage points (pp) (Fig. 1) for the VFMs. The performance decrease on the out-of-domain datasets varies largely depending on the model. Kaiko-B achieves the lowest results on all subdatasets. In comparison, the other two Kaiko models perform similarly and are more robust on the subdatasets. The larger VFMs show the lowest decrease and are superior in terms of robustness. Virchow2 achieves the highest auroc on average on all subdatasets. This might be caused by the focus on pathology-specific augmentations within pre-training. UNI2-h and H-optimus-1 perform similarly on *uke.scanner* and *uke.first*. As *uke.scanner* is not included in the fine-tuning dataset, it might result from the fact that both models were pre-trained with data from multiple scanners. The lowest decrease is found for *uke.spot* which is considered as in-domain dataset since the same imaging protocol was used as for *uke.first*.

Overall, the evaluation shows that while VFMs with more parameters are superior on in-domain datasets and more robust on out-of-domain datasets for prostate cancer relapse prediction, they still show a decrease in performance across subdatasets due to the domain shifts. The reasons for this as well as possible solutions should be further evaluated and addressed in future work.

References

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